Gabriel Eze

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EDUCATION

B.Sc. Data Science & Mathematics, *Centre College* — 08/22 – 05/26 | GPA: 3.44

Relevant Coursework: **Abstract Algebra, Advanced Interpolation Methods, Applied Machine Learning, Bayesian Data Analysis, Building AI-Powered Applications, Data Manipulation & Visualization, Linear Algebra I & II, Mathematical Statistics, Probability Theory, Python Programming and Problem Solving, Real Analysis, Statistical Modeling.**

SKILLS

Computing — Harmonic Analysis, Logic, Q-Ball Imaging

LLM — Chunking, Doc2Query, Embeddings, Hit@K, Prompt Engineering, Red-Team Testing, Tool Use

Programming — MATLAB, Python, R, SQL

Tools — ChromaDB, Endpoints, Git, JupyterLab, LaTeX, MySQL, Pydantic, Quarto, Regex, Tableau, VS Code

RESEARCH

Stack Domination Density of Hypercubes, *Graph Theory* — 05/25 – 08/25 | Dr. Brauch, Dr. Wiglesworth

- Formulated and proved results for stack domination density in $C_m \times P_2 \times P_n$, deriving bounds across residue classes of m.
- Designed modular "Clockwise" and "Domination" algorithms to construct minimal dominating sets and validate proofs through large-scale computation.
- Co-authored a 38-page research manuscript integrating combinatorial proofs and algorithmic verification for potential undergraduate publication.

Pitch Modeling, *Carnegie Mellon SURE Camp* — 06/24 – 07/24 | Quang Nguyen, Dr. Yurko

- Conducted advanced statistical analysis and modeling to predict pitch effectiveness (Stuff+) for MLB players, leveraging large datasets from FanGraphs and Baseball Savant.
- Applied Random Forest, Lasso regression, and HGBR models to analyze the complete arsenal of pitches per pitcher, enhancing predictive accuracy by modeling multiple pitch types and providing competitive RMSE results.
- Developed a pitch recommendation tool that provided actionable insights for MLB teams, optimizing player performance by suggesting ideal pitch sequencing based on past game data.

Sports Analytics — 05/23 – 07/23 | Dr. Heath, Dr. Poudel

- Utilized Python to lead data cleaning and parsing efforts within a complex big data pipeline, ensuring data accuracy and reliability for basketball game analysis.
- Tracked possession data and employed advanced statistical techniques to assess player substitutions, providing vital insights for informed coaching decisions and game outcome analysis.
- Collaborated on the development of win probability models based on real-time game data, empowering the coaching staff with actionable insights for strategic game planning.

PERSONAL PROJECTS

Forest Fire Simulation — 11/22 – 12/22

- Designed class-based forest simulation modeling state transitions on a 10×15 grid.
- Implemented probabilistic burn dynamics and interactive GUI controls.

DNA Sequence Analysis — 10/22 – 11/22

- Developed sliding-window STR detection algorithm to quantify repeat sequence from genomic input.
- Parsed CSV database and programmatically matched repeat counts to identify genetic profiles.

AWARDS

Best Visualization, *ASA DataFest (EKU)* — 2025

Analyzed national office leasing data to deliver strategic market insights for Savills in 3-day competition.